Supplemental Table 1. The inclusion and exclusion criteria in this study

Year, first author	Weight status	Baseline BMI (kg/m^2)		Vitamin D status	Baseline 25OHD (ng/ml)		Included by HbA1c	Supplemental history	Pregnant or lactating
		Intervene	Control		Intervene	Control			
Diabetes									
2019 Lo	NA	37.4 ± 2.4	39.8 ± 2.5	<20 ng/ml	15.1 ± 1.2	14.4 ± 0.9	7.5–10%	<1,000 IU/day	Not pregnant
2018 Upreti	NA	24.6 ± 3.8	25.6 ± 4.2	<30 ng/ml	25.3 ± 8.6	26.7 ± 11.9	≤8.5%	None	None
2018 Riek	NA	36.9 ± 6.8	31.5 ± 1.6	<25 ng/ml	17.3 ± 1.9	17.4 ± 1.4	5.5–9.5%	NA	Not pregnant
2018 Khan	NA	NA	NA	<20 ng/ml	13.5 ± 3.7	13.4 ± 3.4	7.5–9.0%	None in 3 month	NA
2017 Randhawa	NA	23.5 ± 3.7	25.5 ± 4.0	<20 ng/ml	16.5 ± 1.6	17.1 ± 1.7	7.0–9.0%	NA	Not pregnant
2017 Gulseth	BMI≤45	NA	NA	<20 ng/ml	15.2 ± 4.8	14.7 ± 5.0	≤11.0%	No high dose	None
2017 Agarwal	NA	NA	NA	<30 ng/ml	25.6 ± 9.9	29.1 ± 14.9	7.0–8.5%	NA	Not pregnant
2017 Alireza	BMI≥25	30.7 ± 5.1	29.7 ± 4.3	<20 ng/ml	16.9 ± 3.6	16.6 ± 4.6	NA	None in 6 month	NA
2015 Sadiya	BMI≥30	NA	NA	<20 ng/ml	11.4 ± 3.68	12.2 ± 4.5	NA	None	None
2014 Tabesh(1)	NA	30.5 ± 5.3	30.3 ± 3.8	<30 ng/ml	11.2 ± 5.6	18.3 ± 6.6	NA	None	None
2014 Tabesh(2)	NA	29.9 ± 4.5	30.4 ± 4.2	<30 ng/ml	12.2 ± 6.6	22.3 ± 6.0	NA	None	None
2014 Kampmann	NA	35.3 ± 2.9	32.4 ± 2.0	<20 ng/ml	12.4 ± 2.0	13.9 ± 1.5	NA	NA	None
2014 Ryu	NA	NA	NA	<20 ng/ml	10.8 ± 5.1	10.1 ± 3.9	<8.5%	None in 3 month	NA
2014 Baziar	25≤BMI≤45	27.3 ± 1.6	27.3 ± 1.4	10–30 ng/ml	14.3 ± 5.9	15.5 ± 5.6	≤8.0%	None in 6 month	Not pregnant
2013 Yiu	NA	NA	25.1 ± 3.4	<30 ng/ml	21.1 ± 4.4	21.9 ± 4.1	≤11.0%	NA	None
2008 Sugden	NA	31.7 ± 6.4	31.7 ± 6.5	<20 ng/ml	16.1 ± 4.1	14.6 ± 3.4	NA	None	NA
Prediabetes									
2020 Bhatt	BMI≥23	31.1 ± 6.2	28.8 ± 3.9	<30 ng/ml	12.0 ± 2.4	12.8 ± 2.1	NA	None in 6 month	None
2019 Wallace	Yes	34.7 ± 8.0	33.9 ± 5.6	<20 ng/ml	12.3 ± 5.7	12.0 ± 6.2	NA	None	None
2019 Niroomand	NA	31 ± 6	32 ± 6	<30 ng/ml	12.3 ± 6.6	12.7 ± 6.3	NA	None	NA
2017 Moreira	NA	30.1 ± 3.9	31.7 ± 4.9	<26 ng/ml	19.2 ± 5.7	19.0 ± 5.7	5.5–6.4%	NA	None
2016 Wagner	BMI≤32	28.3 ± 3.6	28.6 ± 2.6	<30 ng/ml	16.8 ± 5.9	18.8 ± 3.3	≤7.9%	None	None
2015 Barengolts	28≤BMI≤39	32.4 ± 2.9	31.5 ± 2.4	<30 ng/ml	NA	NA	<7.0%	NA	NA
2014 Dutta	NA	26.3 ± 4.5	26.8 ± 4.6	<30 ng/ml	17.0 ± 7.7	18.0 ± 7.2	NA	None in 6 month	Not pregnant
2014 Oosterwerff	BMI≥27	32.2 ± 4.8	33.2 ± 5.1	<20 ng/ml	10.0 ± 4.3	8.7 ± 4.2	NA	<200 IU/day	None
2013 Davidson	WC≥40/35	32.1 ± 4.7	32.9 ± 4.3	<30 ng/ml	22.0 ± 4.5	22.0 ± 4.8	5.8–7.0%	NA	NA